Algebraic Curves
Chapter 8.2. The Vector Spaces $\mathrm{L}[D]$

Let $D = \sum_{\mathbf{p}} n_{\mathbf{p}}\mathbf{p}$ be an arbitrary divisor on $\mathbb{X}$. We want to determine when there exists and how many there are of rational functions with poles only at chosen points, where the order of the pole is no worse than $n_{\mathbf{p}}$ at $\mathbf{p}$. Define,

$$\mathrm{L}[D] = \{\mathrm{f} \in \mathbb{K} \mid \forall \mathbf{p} \in \mathbb{X} \colon \mathrm{ord}_{\mathbf{p}}[\mathrm{f}] \geq -n_{\mathbf{p}}\}$$

Hence a rational function f is in $\mathrm{L}(D)$ if $\mathrm{div}[\mathrm{f}] + D \geq 0$ or if $\mathrm{f} = 0$. Observe that $\mathrm{L}[D]$ forms a vector space over $\Bbbk$. Let its dimension be denoted $\ell[D]$.

Proposition.
We have the following,

1. If $D \leq D'$, then $\mathrm{L}[D] \subseteq \mathrm{L}[D']$ and $\dim_{\Bbbk} \mathrm{L}[D']/\mathrm{L}[D] \leq \deg[D' - D]$
2. $\mathrm{L}(0) = \Bbbk$. $\mathrm{L}[D] = 0$ if $\deg[D] < 0$.
3. $\mathrm{L}[D]$ is finite-dimensional for all D. If $\deg D \geq 0$, then $\ell[D] \leq \deg D + 1$.
4. If $D \sim D'$, then $\ell[D] = \ell[D']$

Proof.

For the first proposition write,

$$D' = D + 1\mathbf{p}_1 + \cdots + 1\mathbf{p}_s$$

It is thus immediate,

$$\mathrm{L}[D] \subseteq \mathrm{L}[D + 1\mathbf{p}_1] \subseteq \cdots \subseteq \mathrm{L}[D + 1\mathbf{p}_1 + \cdots + 1\mathbf{p}_s]$$

Moreover by induction it suffices to show $\dim \mathrm{L}[D + 1\mathbf{p}]/\mathrm{L}[D] \leq 1$. For this let t be the uniformizing parameter of $\mathscr{O}_{\mathbf{p}}[\mathbb{X}]$. Define,

$$\varphi \colon \mathrm{L}[D + 1\mathbf{p}] \to \Bbbk, \qquad \mathrm{f} \mapsto \mathrm{eval}_{\mathbf{p}}\left[t^{n_{\mathbf{p}}+1}\,\mathrm{f}\right]$$

Given $\mathrm{ord}_{\mathbf{p}}[\mathrm{f}] \geq -n_{\mathbf{p}} - 1$ this is well-defined. Moreover φ is a linear map with $\ker \varphi = \mathrm{L}[D]$, so induces a linear map $\bar{\varphi} \colon \mathrm{L}[D + 1\mathbf{p}]/\mathrm{L}[D] \to \Bbbk$. The result then follows by the First Isomorphism Theorem.

The second proposition follows by Chapter 8.1. Corollary 1 and Chapter 8.1. Proposition 2.

For the third proposition let $\deg D = n \geq 0$ and choose $\mathbf{p} \in \mathbb{X}$. Let $D' = D - (n+1)\mathbf{p}$. Hence $\mathrm{L}[D'] = 0$ and from the first point $\dim \mathrm{L}[D]/\mathrm{L}[D'] \leq n+1$, so $\ell[D] \leq n+1$.

For the fourth proposition let $D' = D + \mathrm{div}[\mathrm{g}]$ for some g. Define,

$$\psi\colon \mathrm{L}[D] \to \mathrm{L}[D'], \qquad \mathrm{f} \mapsto \mathrm{f}\,\mathrm{g}$$

Given ψ is an isomorphism of vector spaces, $\ell[D] = \ell[D']$.

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For any subset $S \subseteq \mathbb{X}$ and divisor $D = \sum_{\mathbf{p}} n_{\mathbf{p}}\mathbf{p}$, define,

$$\deg^S[D] = \sum_{\mathbf{p}\in S} n_{\mathbf{p}}, \qquad \mathrm{L}^S[D] = \{\mathrm{f} \in \mathbb{K} \mid \forall \mathbf{p} \in S\colon \mathrm{ord}_{\mathbf{p}}[\mathrm{f}] \geq -n_{\mathbf{p}}\}$$

The previous results yield the following lemma,

Proposition.
If $D \leq D'$, then $\mathrm{L}^S[D] \subseteq \mathrm{L}^S[D']$. Moreover if S is finite, then $\dim \mathrm{L}^S[D']/\mathrm{L}^S[D] = \deg^S[D' - D]$.

Proof.

By induction it suffices to prove in the case $D' = D + 1\mathbf{p}$. Define φ the same way as before. As S is finite there exists $\mathrm{f} \in \mathbb{K}$ s.t. $\mathrm{ord}_{\mathbf{p}}[\mathrm{f}] = -n_{\mathbf{p}} - 1$ and $\mathrm{ord}_{\mathbf{q}}[\mathrm{f}] \geq -n_{\mathbf{q}}$ for all other $\mathbf{q} \in S$. Hence φ maps $\mathrm{L}^S[D+1\mathbf{p}]$ onto $\Bbbk$ as an isomorphism. The result then follows by the First Isomorphism Theorem.

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The following result is our main theorem.

Proposition.
Let $\mathrm{x} \in \mathbb{K}$ and not unit. Let $n = [\mathbb{K} : \Bbbk(\mathrm{x})]$. Then $(\mathrm{x})_0$ is an effective divisor of degree n and there exists a constant τ s.t. $\ell[r(\mathrm{x})_0] \geq rn - \tau$ for all r.

Proof.

Write $Z := Z = \sum_{\mathbf{p}} n_{\mathbf{p}}\mathbf{p}$ and let m be its degree. We first show $m \leq n$. Let $S = \{\mathbf{p} \in \mathbb{X} \mid n_{\mathbf{p}} > 0\}$. By Lemma 1 we may choose $\mathrm{v}_1, \ldots, \mathrm{v}_m \in \mathrm{L}^S[0]$ s.t. their residues $\bar{\mathrm{v}}_1, \ldots, \bar{\mathrm{v}}_m \in \mathrm{L}^S[0]/\mathrm{L}^S[-Z]$ are a basis for this space. We claim they are linearly independent over $\Bbbk(\mathrm{x})$. Assume for contradiction they were not. Then by clearing denominators and multiplying by powers of x there exists polynomials $\mathrm{g}_i = \lambda_i + \mathrm{x}\,\mathrm{h}_i \in \Bbbk[\mathrm{x}]$ with $\lambda_i \in \Bbbk$ and $\sum_i \mathrm{g}_i\,\mathrm{v}_i = 0$ where not all λ_i are zero. However then,

$$\sum_i \lambda_i\,\mathrm{v}_i = -\mathrm{x}\sum_i \mathrm{h}_i\,\mathrm{v}_i \in \mathrm{L}^S[-Z]$$

So $\sum_i \lambda_i \bar{\mathrm{v}}_i = 0$, a contradiction. Therefore $m \leq n$.

We next prove the second proposition before returning to the first. Let $\mathrm{w}_1, \ldots, \mathrm{w}_n$ be the basis of $\mathbb{K}$ over $\Bbbk(\mathrm{x})$. By Question 1.56 each w_i satisfies an equation,

$$\mathrm{w}_i^{n_i} + \mathrm{a}_{i,1}\, \mathrm{w}_i^{n_i - 1} + \cdots = 0$$

Where $\mathrm{a}_{i,j} \in \Bbbk[\mathrm{x}^{-1}]$. Hence $\mathrm{ord}_{\mathbf{p}} \geq 0$ if $\mathbf{p}[\mathrm{w}_i^{n_i}] \notin S$. Otherwise $\mathrm{ord}_{\mathbf{p}}[\mathrm{w}_i] < 0$ and $\mathbf{p} \notin S$ implies $\mathrm{ord}_{\mathbf{p}}[\mathrm{w}_i^{n_i}] < \mathrm{ord}_{\mathbf{p}}[\mathrm{a}_{i,j}\, \mathrm{w}_i^{n_i - j}]$, a contradiction. Thus for some $t > 0$, $\mathrm{div}[\mathrm{w}_i] + tZ \geq 0$ for all $i \in \{0, \ldots, n\}$. Hence $\mathrm{w}\, \mathrm{x}^{-j} \in \mathrm{L}[(r+t)Z]$ for all $i \in \{0, \ldots, n\}$ and $j \in \{0, \ldots, r\}$. As the set of w_i are linearly independent over $\Bbbk(\mathrm{x})$ and $1, \ldots, x^{-r}$ are linearly independent over $\Bbbk$, hence the element-wise product of these two sets is linearly independent over $\Bbbk$. Thus $\ell[(r+t)Z] \geq n(r+1)$. However by the previous proposition,

$$\ell[(r+t)Z] = \ell[rZ] + \dim_{\Bbbk} \mathrm{L}[(r+t)Z]/\mathrm{L}[rZ] \leq \ell[rZ] + tm$$

Therefore $\ell[rZ] \geq n(r+1) - tm = rn - \tau$ as desired.

Thus, as $rn - \tau \leq \ell(rZ) \leq rm + 1$, for sufficiently large r: $n \leq m$.

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As corollaries to the previous result the following are equivalent,

1. $\mathbb{C}$ is rational
2. $\mathbb{X}$ is isomorphic to $\mathbb{P}^1$
3. There exists $\mathrm{x} \in \mathbb{K}$ s.t. $\deg\,(\mathrm{x})_0 = 1$
4. There exists $\mathbf{p} \in \mathbb{X}$ s.t. $\ell[1\mathbf{p}] > 1$

Question 8.8

Proposition.
If $D \leq D'$, then $\ell[D'] \leq \ell[D] + \deg[D' - D]$.

Proof.

We prove $\dim_{\Bbbk} \mathrm{L}[D']/\mathrm{L}[D] = \ell[D'] - \ell[D]$. This is a standard result of linear algebra. Given $\mathrm{L}[D']$ and $\mathrm{L}[D]$ are both vector spaces with the former a superspace of the latter, there exists a basis $\mathbf{e}_1, \ldots, \mathbf{e}_{\ell[D]}$ of $\mathrm{L}[D]$ and an extension of this set $\mathbf{e}_1, \ldots, \mathbf{e}_{\ell[D']}$ to a basis of $\mathrm{L}[D']$. By module-quotient the basis of $\mathrm{L}[D']/\mathrm{L}[D]$ is then precisely $\mathbf{e}_{\ell[D]+1}, \ldots, \mathbf{e}_{\ell[D']}$, so this is a vector space of dimension $\ell[D'] - \ell[D]$. Therefore,

$$\ell[D'] - \ell[D] = \dim_{\Bbbk} \mathrm{L}[D']/\mathrm{L}[D] \leq \deg[D' - D]$$

At which our proposition follows.

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Question 8.9

Proposition.
Let $\mathbb{X} = \mathbb{P}^1$. Then $\ell[r(t)_0] = r + 1$.

Proof.

Recall that $t = X_1/X_2$, so that $(t)_0 = X_1$. Namely $\operatorname{div}[X_1] = 1[0:1]$, so that $r \operatorname{div}[X_1] = r[0:1]$. Hence $\mathrm{L}[r(t)_0] = \mathrm{L}[r \operatorname{div} X_1]$ is precisely the set of rational polynomials for which their pole order at $[0:1]$ is at most r. Equivalently, $\operatorname{div}[\mathrm{z}] + r[0:1] \geq 0$. We claim that $\{1, t^{-1}, \ldots, t^{-r}\}$ forms a basis of $\mathrm{L}[r(t)_0]$. The 1 in this set is to represent $\Bbbk$. For this observe that the generic element z of $\mathrm{L}[r(t)_0]$ is s.t. it may be factored about $[0:1]$ to yield, using the uniformizing parameter,

$$\mathrm{z} = \mathrm{a}\, t^{-s}$$

Where $s \leq r$. As this is necessarily the only pole of z, a has only zeroes, i.e. is globally regular, so must be constant $\mathrm{a} = a$. This is precisely our basis. Thus we have the complete characterization,

$$\mathrm{L}[r(t)_0] = \{at^{-s} \mid s \leq r, a \in \Bbbk\}$$

And $\ell[r(t)_0] = r + 1$ as desired.

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Question 8.10

Proposition.
Let $\mathbb{X} = \mathbb{C}$ be a nonsingular cubic and x, y be as in Question 8.2. Define $z = x^{-1}$. Then $\mathrm{L}[r(z)_0] \subseteq \Bbbk[x, y]$ and $\ell[r(z)_0] = 2r$ if $r > 0$.

Proof.

W.l.o.g. by a projective automorphism let $\mathbb{X} = \mathbb{C} = \{Y^2Z - X(X - Z)(X - \lambda Z) = 0\}$ for $\lambda \notin \{0, 1\}$. Recall that $x = X/Z$ and $y = Y/Z$ with,

$$\mathrm{div}[x] = \mathrm{div}[X/Z] = -2[0:1:0] + 2[0:0:1]$$

And so $(z)_0 = 2[0:1:0]$. Thus $\mathrm{L}[r(z)_0]$ are precisely rational functions $\mathrm{v} \in \mathbb{K}$ s.t. $\mathrm{div}[\mathrm{z}] + 2r[0:1:0] \geq 0$. Given that therefore $[0:1:0]$ may be the only pole of v, it has the form,

$$\mathrm{div}[\mathrm{v}] = -s[0:1:0] + \sum_{\mathbf{p}} \lambda_{\mathbf{p}} \mathbf{p}$$

Where $\sum_{\mathbf{p}} \lambda_{\mathbf{p}} = s$ and $\mathbf{p}$ may be located arbitrarily on $\mathbb{C}$. However observe that when $Z = 0$ on $\mathbb{C}$, necessarily $X = 0$, so that the only point with these properties is $[0:1:0]$, where we know v has a root. At any other point we are able to homogenize w.r.t Z, i.e.,

$$\mathrm{v}[X, Y, Z] = \mathrm{a}\, Z^s\, \mathrm{b}[X/Z, Y/Z, 1] = \mathrm{a}\, Z^s\, \mathrm{b}_*[x, y]$$

Where a and b contains the poles and zeroes of v respectively, and $s = \deg \mathrm{v} \leq 2r$. We know that the pole is necessarily at $[0:1:0]$. About here the uniformizing parameter is X/Y so we may locally factor,

$$\mathrm{v} = \left(\frac{Y}{X}\right)^s (1 + \text{h.o.t.})$$

The initial term describes the pole at $[0:1:0]$. As we know from the form of v that a is the term which describes this, we see that a is precisely this term. Hence,

$$\mathrm{v}[X, Y, Z] = \left(\frac{Y}{X}\right)^s Z^s\, \mathrm{b}_*[x, y]$$

Therefore homogenizing in Z, i.e. $X \to X/Z = x, Y \to Y/Z = y$ yields,

$$\mathrm{v}_*[x, y] = \left(\frac{y}{x}\right)^s \mathrm{b}_*[x, y]$$

To show that this belongs to $\Bbbk[x, y]$ it suffices to show it has no pole. Evidently it may only when $x = 0$, but then $y = 0$, so the pole would have to be at $[0:0:1]$. However being $\mathrm{L}[r(z)_0]$ the only place where v is allowed to have a pole is $[0:1:0]$, and we see that this not such a point, so that this must be cancelled out by b_*. Hence v_* has no pole, so is indeed in $\Bbbk[x, y]$.

For the second proposition note that the above implies that the generic element of $\mathrm{L}[r(z)_0]$ is of the form,

$$\mathrm{v}_*[x, y] = x^{-s} y^s \sum_{i+j \leq s} \lambda_{i,j} x^i y^j$$

With the condition that its pole order at $[0 : 1 : 0]$ be greater-than-or-equal to $2r$. Now here $\mathrm{ord}[x] = -2$ and $\mathrm{ord}[y] = -3$, and $y^2 = x^3 + \cdots$, so that all terms may be written as $x^s y$ or x^s for degree $s \geq 0$. Imposing our pole condition is therefore that $-2r \leq -2s - 3$ or $-2r \leq -2s$. The first condition yields $r \geq s + 1.5$, so there are $r - 1$ choices, and the second condition yields $r \leq s$ so $r + 1$ options. Therefore in total there are $2r$ options, so indeed $\ell[r(z)_0] = 2r$.

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Question 8.11

Proposition.
Let D be a divisor. Then $\ell[D] > 0$ *iff* D is linearly equivalent to an effective divisor.

Proof.

We prove the contrapositive. Let D not be linearly equivalent to an effective divisor, so that $D + \mathrm{div}[z] \ngeq 0$ for all $z \in \mathbb{K}$. Therefore $\mathrm{L}[D] = \{0\}$ so $\ell[D] = 0$. Now let $\ell[D] = 0$, so that no matter what $z \in \mathbb{K}$, we have $D + \mathrm{div}[z] \ngeq 0$. Therefore D is not linearly equivalent to an effective divisor.

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Question 8.12

Proposition.
$\deg D = 0$ and $\ell[D] > 0$ *iff* $D \sim 0$

Proof.

Backwards implication is immediate. For forwards implication assume for contradiction $D \nsim 0$, so that $D \neq \operatorname{div}[z]$ for some $z \in \mathbb{K}$, i.e. is not the divisor of a rational function. Moreover $0 < \ell[D] \leq \deg D + 1 = 1$ so that $\ell[D] = 1$, i.e.,

$$\mathrm{L}[D] = \operatorname{span}\{\operatorname{div}[z]\}$$

For some $z \in \mathbb{K}$. Namely $D + k \operatorname{div}[z] \geq 0$ for all $k \in \Bbbk$. Let $\sum_{\mathbf{p}} n_{\mathbf{p}} \mathbf{p}$ be the poles of D, so that this statement means that the zeroes $\sum_{\mathbf{q}} m_{\mathbf{q}} \mathbf{q}$ is also present at all $\mathbf{p}$, and each here $m_{\mathbf{p}} \geq n_{\mathbf{p}}$. Moreover at the zeroes $\sum_{\mathbf{p}'} n_{\mathbf{p}'} \mathbf{p}'$ of D we also have that the poles $\sum_{\mathbf{q}'} m_{\mathbf{q}'} \mathbf{q}'$ are also prevalent and of lesser order $n_{\mathbf{p}'} \geq m_{\mathbf{p}'}$. However both divisors are of degree zero so that,

$$\sum_{\mathbf{p}} n_{\mathbf{p}} - \sum_{\mathbf{p}'} n_{\mathbf{p}'} = 0, \qquad \sum_{\mathbf{p}} m_{\mathbf{p}} - \sum_{\mathbf{p}'} m_{\mathbf{p}'} + \sum_{\mathbf{q}} m_{\mathbf{q}} = 0$$

In particular each $m_{\mathbf{q}}$ must be nonnegative, or else $D + \operatorname{div}[z]$ would be negative there. However from previous we know that the zeroes of D are greater than the poles of $\operatorname{div}[z]$, but also that the poles of $\operatorname{div}[z]$ are of lesser magnitude than the zeroes of D, i.e.

$$\sum_{\mathbf{p}'} m_{\mathbf{p}'} \leq \sum_{\mathbf{p}'} n_{\mathbf{p}'} = \sum_{\mathbf{p}} n_{\mathbf{p}} \leq \sum_{\mathbf{p}} m_{\mathbf{p}}$$

So that $\sum_{\mathbf{p}} m_{\mathbf{p}} - \sum_{\mathbf{p}'} m_{\mathbf{p}'} \geq 0$, and therefore for degree zero each $m_{\mathbf{q}}$ must be zero and $n_{\mathbf{p}} = m_{\mathbf{p}}, n_{\mathbf{p}'} = m_{\mathbf{p}'}$. But then $D = \operatorname{div}[z]$, a contradiction.

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Question 8.13

Proposition.
Let $\ell[D] > 0$ and $0 \neq \mathrm{f} \in \mathrm{L}[D]$. Then $\mathrm{f} \notin \mathrm{L}[D - 1\mathbf{p}]$ for all but a limited number of $\mathbf{p}$.

Proof.

Assume for contradiction $\mathrm{f} \in \mathrm{L}[D - 1\mathbf{p}]$ for all $\mathbf{p}$. The condition $\mathrm{f} \in \mathrm{L}[D - 1\mathbf{p}]$ means that $D + \mathrm{div}[\mathrm{f}] - 1\mathbf{p} \geq 0$. That this holds for all $\mathbf{p}$ implies that $D + \mathrm{div}[\mathrm{f}]$ must have a positive component at all $\mathbf{p}$, so that the difference with $1\mathbf{p}$ is still positive. However D and $\mathrm{div}[\mathrm{f}]$ are divisors, so have a limited number of positive components, so $D + \mathrm{div}[\mathrm{f}]$ has a limited number of positive components. This is a contradiction. Therefore f cannot be in $\mathrm{L}[D - 1\mathbf{p}]$ for all $\mathbf{p}$, but may be in so for but only a limited number of $\mathbf{p}$.

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